Sharp asymptotic separation of opposing egalitarian core selections

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Abstract

For singleton-normalized exact transferable-utility games on n players, let s_n denote the supremum of the normalized ℓ_1 distance between the leximin core allocation and the core allocation that lexicographically minimizes decreasingly sorted payoffs. We prove

$$\lim_{n\to\infty} n(2-s_n) = 2, \qquad 2 - s_n = \frac{2}{n} + O\left(\frac{\log n}{n^2}\right).$$

The universal bound holds for opposing maximin and minimax points in every compact convex nonnegative constant-total set. For the matching asymptotic construction, we realize a line segment as the entire core of the minimum of two additive games. Complementary coalition equalities encode a Fibonacci recurrence. Both full selectors are uniquely determined at their first optimization stages. The construction exists for every sufficiently large n and admits integer generators with polynomially bounded total. The argument is self-contained; finite exact computations are supplied as reproducibility checks.

1 The question and the main result

The core imposes coalition stability but often leaves several efficient allocations available. Two egalitarian refinements compare those allocations in opposite orders: leximin first improves the poorest payoff, whereas the opposing selection first lowers the richest payoff. How far apart can their recommendations be, even when every coalition value is attained somewhere in the core?

Let N be a finite player set, $n = |N|$, and let a TU game be a function $v : 2^N \to \mathbb{R}$ with $v(\varnothing) = 0$. Write $y(S) = \sum_{i\in S} y_i$ and

$$\mathrm{Core}(v) = \{y \in \mathbb{R}^N : y(N) = v(N),\ y(S) \geq v(S) \text{ for all } S \subseteq N\}.$$

The game is *exact* if its core is nonempty and, for every coalition S, some core allocation y satisfies $y(S) = v(S)$. Throughout we impose *singleton normalization*, $v(\{i\}) = 0$ for all i, and a positive grand total $T = v(N)$. Its core is then a compact convex subset of $\{y \geq 0 : y(N) = T\}$.

For $y \in \mathbb{R}^N$, denote its increasingly and decreasingly sorted vectors by $y^{\uparrow}$ and $y^{\downarrow}$. We use the following explicit conventions:

$$x = \mathrm{Lmin}(v): \quad x^{\uparrow} \text{ is lexicographically maximal over } \mathrm{Core}(v),$$

$$z = \mathrm{Lmax}(v): \quad z^{\downarrow} \text{ is lexicographically minimal over } \mathrm{Core}(v).$$

Thus Lmax means the lexicographic refinement of *minimizing* the maximum payoff. Both rules sort all individual payoffs. Neither rule here sorts coalition excesses.

These full selections exist and are unique on a nonempty compact convex set. Existence follows by successive optimization of continuous order statistics on compact sets. For uniqueness of the first rule, two distinct optimal points would have the same sorted vector. Consider the smallest payoff level whose assigned player sets differ. Their midpoint keeps all lower levels unchanged and has fewer coordinates at this level, strictly improving the sorted vector. Convexity gives a contradiction. Applying the same argument to the negative of the feasible set proves uniqueness of the second rule.

Put

$$O(x,z)=\sum_{i\in N}\min(x_i,z_i),\qquad \rho(v)=\frac{\|x-z\|_1}{T}=2-\frac{2O(x,z)}{T},\qquad s_n=\sup_v \rho(v), \tag{1}$$

where the supremum ranges over all singleton-normalized exact games with n players and positive total. Scaling the game by a positive constant leaves ρ unchanged. No attainment of this supremum is assumed.

Theorem 1. *For every $n\ge 2$,*

$$s_n\le 2-\frac{2}{n-1}.$$

For every integer $n\ge 128$, let $L=\lceil \log_2 n\rceil$. Then

$$\frac{2}{n-1}\le 2-s_n\le \frac{2}{n-12L}+\frac{2}{n^2+1}. \tag{2}$$

Consequently

$$\lim_{n\to\infty} n(2-s_n)=2,\qquad 2-s_n=\frac{2}{n}+O\left(\frac{\log n}{n^2}\right). \tag{3}$$

The lower construction for s_n uses two nonnegative efficient generators and has a one-dimensional entire core. It can be scaled to an integer-valued game; an alternative parameter choice gives integer generator coordinates of $O(\log n)$ bits and total $O(n^3)$.

The normalized vectors x/T and z/T are probability vectors, so $\rho/2$ is their total variation distance. The result concerns disagreement about distributing a fixed total: it does not assert a loss of aggregate worth. Fixed-n sharpness, an exact second-order term, and uniqueness of asymptotic extremizers are not asserted.

Relation to prior work. The leximin stable allocation in the core is established in Arin and Iñarra [1]. Vazirani [2], notably Sections 1.2 and 5, studies the two opposing fairness criteria for assignment-game cores and gives efficient algorithms; his Lemma 3 also proves uniqueness of leximin. Gangam et al. [3] study corresponding algorithms for max-flow, MST, and b-matching games, with their constructive algorithms operating on Owen sets. These domains and algorithmic objectives differ from the unrestricted exact-game distance supremum considered here.

Exactness and lower-envelope representations are classical. For example, Amarante et al. [4], Section 2.2, record both the lower-envelope interpretation and the fact that its core may strictly contain the generating convex hull. Studený and Kratochvíl [5] give a facet description of the finite exact-game cone. Our proof uses neither a facet enumeration nor an unproved replacement of the core by its generators. The proposed contribution is the quantitative asymptotic theorem and the explicit whole-core realization below. A bounded source search did not identify an equivalent result; specialist assessment of priority and significance remains pending. The companion source ledger records the versions and access limitations of that search.

2 A universal overlap inequality

Lemma 2. *Let $n \geq 2$, $T > 0$, and let K be a nonempty compact convex subset of $\{y \in \mathbb{R}^n : y \geq 0,\ y(N) = T\}$. If*

$$x \in \arg\max_{y \in K} \min_i y_i, \qquad z \in \arg\min_{y \in K} \max_i y_i,$$

then $\sum_i \min(x_i, z_i) \geq T/(n-1)$.

Proof. If $x = z$, the overlap is T. Otherwise let $a = \min_i x_i$ and $M = \max_i z_i$. Since $a \leq T/n \leq M$, equality $a = M$ would force both points to be uniform; hence $a < M$.

Some index i with $x_i = a$ satisfies $z_i \leq a$. If every minimum coordinate instead had $z_i > a$, a sufficiently small positive move from x toward z would raise all of them, while all other coordinates would remain above a. This contradicts maximin optimality. Similarly some index j with $z_j = M$ has $x_j \geq M$: otherwise a small move from z toward x lowers all maximum coordinates and keeps the others below M. The two indices are distinct because $x_i = a < M \leq x_j$. Therefore

$$O(x, z) \geq z_i + M,$$

and nonnegativity gives

$$T \leq z_i + (n-1)M \leq (n-1)(z_i + M) \leq (n-1)O(x, z).$$

□

By (1), the lemma proves the upper bound on s_n in Theorem 1. Exactness and all later lexicographic stages are unnecessary for this bound.

3 Realizing the entire core as a segment

Lemma 3. *Let $p \geq 1$ and let $A \in \{0,1\}^{p \times p}$ be invertible with $d = A^{-1}\mathbf{1} > 0$. Choose an integer $r \geq 1$ such that $H = r - \sum_i d_i \geq 1$. On one distinguished donor, p other donors, and r receivers, define*

$$U = (H, d_1, \ldots, d_p; 0^r), \qquad V = (0^{p+1}; 1^r), \qquad v(S) = \min\{U(S), V(S)\}.$$

Then v is singleton-normalized and exact, has total r, and $\mathrm{Core}(v) = [U, V]$. If $\alpha = \min_i d_i$, its full selectors equal $tU + (1-t)V$ at

$$t_{\min} = \frac{1}{1+\alpha}, \qquad t_{\max} = \frac{1}{1+H}. \tag{4}$$

In particular,

$$2 - \rho(v) = 2\left(\frac{\alpha}{1+\alpha} + \frac{1}{1+H}\right). \tag{5}$$

Proof. Both generators are nonnegative, have total r, and satisfy all coalition lower bounds. Every coalition value is attained by one of them, establishing exactness. Their supports are disjoint, establishing singleton normalization. Since every column of A is nonzero, positivity and $Ad = \mathbf{1}$ imply $d_i \leq 1$ for every i.

Let D be the subset of nondistinguished donors specified by any row of A, and let e be any receiver. Both generators give $D \cup \{e\}$ value one, and give its complement value $r - 1$. Thus efficiency and these two coalition inequalities force, at every core point,

$$y(D) + y_e = 1.$$

Fixing a row and varying e shows that all receivers have a common coordinate, say $1-t$. Varying the row gives $Ay_d = t\mathbf{1}$, whence $y_d = td$. Efficiency gives $y_h = tH$ for the distinguished donor. Nonnegativity now implies $0 \le t \le 1$. Conversely every point of $[U, V]$ is in the core by convexity. This proves the whole-core equality.

On the segment the smallest and largest coordinates are

$$\min\{t\alpha, 1-t\}, \qquad \max\{tH, 1-t\},$$

because $0 < \alpha \le d_i \le 1 \le H$. Each has a unique relevant scalar optimum, giving (4). These unique first-stage optima determine the full lexicographic selectors. Since $\|U - V\|_1 = 2r$ and $t_{\min} \ge t_{\max}$,

$$\rho(v) = 2(t_{\min} - t_{\max}),$$

which rearranges to (5). □

The essential restriction is the system of unit-weight donor subsets. Merely defining a game as a lower envelope proves exactness but does not generally prove that its core is the generating segment.

4 The Fibonacci construction and all player counts

Let $F_1 = F_2 = 1$ and $F_{j+2} = F_{j+1} + F_j$. For integers $m \ge 2$ and $r \ge m+1$, set

$$F = F_{m+1}, \quad H = r - m, \quad a_i = F_{m-i+1}/F, \quad b_i = 1 - a_i \quad (1 \le i \le m).$$

Use donor labels $h, A_1, \dots, A_m, B_1, \dots, B_m$ and r receivers, and define

$$U = (H, a_1, \dots, a_m, b_1, \dots, b_m; 0^r), \qquad V = (0^{2m+1}; 1^r). \tag{6}$$

Thus $n = r + 2m + 1$ and $T = r$.

Proposition 4. *The game $v(S) = \min\{U(S), V(S)\}$ from (6) is singleton-normalized and exact, its entire core is $[U, V]$, and*

$$\rho(v) = 2\left(\frac{F}{F+1} - \frac{1}{r-m+1}\right), \qquad 2 - \rho(v) = \frac{2}{F_{m+1}+1} + \frac{2}{r-m+1}. \tag{7}$$

Proof. The following $2m$ subsets of the nondistinguished donors have U-weight one:

$$\begin{aligned} &\{A_i, B_i\} && (1 \le i \le m),\\ &\{A_{i+1}, A_{i+2}, B_i\} && (1 \le i \le m-2),\\ &\{A_{m-1}, B_m\}, \qquad \{A_1, A_2\}. \end{aligned}$$

The pair rows are immediate; the other equalities are the recurrence and the identities $a_{m-1} = a_m$ and $a_1 + a_2 = 1$.

Their incidence matrix is invertible. In its homogeneous system the pair rows eliminate $B_i = -A_i$. The other rows then give

$$A_i = A_{i+1} + A_{i+2}, \qquad A_{m-1} = A_m, \qquad A_1 + A_2 = 0.$$

Writing $A_m = u$ gives $A_i = F_{m-i+1}u$ and $F_{m+1}u = 0$, so every coordinate vanishes. This also covers $m = 2$, when recurrence rows are absent. The positive inhomogeneous solution (a, b) has total m and minimum $1/F$: the smallest a_i is $1/F$, while $\min_i b_i = F_{m-1}/F \ge 1/F$. Apply Lemma 3 with $p = 2m$, $\alpha = 1/F$, and $H = r - m$. □

Proof of Theorem 1. Lemma 2 gives the left inequality in (2). For its right inequality choose, for every $n \geq 128$,

$$L = \lceil \log_2 n \rceil, \qquad m = 4L, \qquad r = n - 8L - 1.$$

We need $r \geq m+1$, equivalently $n \geq 12L+2$. At $n = 128$ this is a direct check. For $L \geq 8$, the least possible n is $2^{L-1}+1$; $2^{L-1}+1 \geq 12L+2$ holds at $L = 8$ and its left side grows faster thereafter. Also $F_{2j+1} \geq 2^j$ for $j \geq 0$, by induction from $F_{2j+3} \geq 2F_{2j+1}$. Hence

$$F_{m+1} \geq 2^{2L} \geq n^2, \qquad r - m + 1 = n - 12L > 0.$$

Proposition 4 proves the required upper bound on the deficit for this exact game. Taking the supremum of separation gives (2). Finally,

$$\frac{2}{n - 12L} - \frac{2}{n} = \frac{24L}{n(n - 12L)},$$

and $L = O(\log n)$, proving (3) by squeezing.

Multiplication by F makes every generator coordinate integral. For the polynomial-total claim choose instead the least $m \geq 2$ with $F_{m+1} \geq n^2$ and again put $r = n - 2m - 1$. For sufficiently large n, $m = O(\log n)$, $r \geq m+1$, and $n^2 \leq F_{m+1} < 2n^2$. The scaled total is $rF_{m+1} = O(n^3)$, and every nonnegative coordinate is bounded by that total. Thus each coordinate uses $O(\log n)$ bits. The two vectors are a compact representation of the game; listing all coalition values would still require exponentially many entries. □

5 Verification and scope of the evidence

The proof above does not depend on a computational census. The companion code reconstructs every proper coalition inequality directly from the two rational generators. Run `run_checks.py` in the `verification` directory to check the cases

$$(m, r) \in \{(2,3), (2,4), (3,4), (3,5), (4,5)\}.$$

These cases have $8, 9, 11, 12, 14$ players. Exact rational Z3 queries exclude every off-segment core allocation, compute both first-stage optimum values, and exclude every other point on either optimal face. Generator feasibility and convexity establish the opposite core containment algebraically. Arithmetic checks also cover $2 \leq m \leq 64$ and displayed large-n parameter choices.

A deliberate perturbation of one donor pair at $(m, r) = (2, 3)$ preserves the common generator total and singleton normalization but breaks the unit-sum realization. The verifier finds and checks an explicit core point outside the proposed segment. This guards against treating every lower envelope as the convex hull of its generators. The code uses explicit exceptions, rejects UNKNOWN and timeouts, and has been replayed with and without Python optimization. Its JSON records are finite corroboration; they are not a formal proof of the all-parameter theorem. Specialist correctness and priority review remain pending.

References

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